Solution

Using equation (10.8)

$v = \omega\sqrt{A^2 - x^2} \Rightarrow v^2 = \omega^2 (A^2 - x^2)$

Therefore, at position x_1,

$$v_1^2 = \omega^2(A^2 - x_1^2) \quad (1)$$

Similarly, at position x_2,

$$v_2^2 = \omega^2(A^2 - x_2^2) \quad (2)$$

Subtrating (2) from (1), we get

$$v_1^2 - v_2^2 = \omega^2(A^2 - x_1^2) - \omega^2(A^2 - x_2^2)$$

$$= \omega^2(x_2^2 - x_1^2)$$

$$\omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}} \Rightarrow T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}} \quad (3)$$

Dividing (1) and (2), we get

$$\frac{v_1^2}{v_2^2} = \frac{\omega^2\left(A^2 - x_1^2\right)}{\omega^2\left(A^2 - x_2^2\right)} \Rightarrow A = \sqrt{\frac{v_1^2 x_2^2 - v_2^2 x_1^2}{v_1^2 - v_2^2}} \quad (4)$$

Dividing equation (3) and equation (4), we have

$$\frac{T}{A} = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 x_2^2 - v_2^2 x_1^2}}$$

10.2.3 Time period, frequency, phase, phase difference and epoch in SHM.

(i) Time period

The time period is defined as the time taken by a particle to complete one oscillation. It is usually denoted by T. For one complete revolution, the time taken is t = T, therefore

$$\omega T = 2\pi \Rightarrow T = \frac{2\pi}{\omega} \quad (10.11)$$

Then, the displacement of a particle executing simple harmonic motion can be written either as sine function or cosine function.

$$y(t) = A \sin \frac{2\pi}{T} t \text{ or } y(t) = A \cos \frac{2\pi}{T} t$$

where T represents the time period. Suppose the time t is replaced by $t + T$, then the function

$$y(t + T) = \text{A} \sin \frac{2\pi}{T} (t + T)$$

$$= A \sin(\frac{2\pi}{T} t + 2\pi)$$

$$= A \sin \frac{2\pi}{T} t = y(t)$$

$$y(t + T) = y(t)$$

Thus, the function repeats after one time period. This $y(t)$ is an example of periodic function.

(ii) Frequency and angular frequency

The number of oscillations produced by the particle per second is called frequency. It is denoted by f. SI unit for frequency is s^{-1} or hertz (In symbol, Hz).

Mathematically, frequency is related to time period by

$$f = \frac{1}{T} \quad (10.12)$$

The number of cycles (or revolutions) per second is called angular frequency. It is usually denoted by the Greek small letter 'omega', ω.

Comparing equation (10.11) and equation (10.12), angular frequency and frequency are related by

$$\omega = 2\pi f \quad (10.13)$$

SI unit for angular frequency is rad s^{-1}. (read it as radian per second)

(iii) Phase

The phase of a vibrating particle at any instant completely specifies the state of the particle. It expresses the position and direction of motion of the particle at that instant with respect to its mean position (Figure 10.11).

$$y = A \sin (\omega t + \varphi_0) \quad (10.14)$$

where $\omega t + \varphi_0 = \varphi$ *is called the phase of the vibrating particle. At time* $t = 0\,s$ *(initial time), the phase* $\varphi = \varphi_0$ *is called epoch (initial phase) where* φ_0 *is called the angle of epoch.*

Phase difference: Consider two particles executing simple harmonic motions. Their

UNIT-10(XI-Physics_Vol-2).indd 196 25-03-2022 20:29:56